

ASEN 6044

Homework #2

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Problem 1:**Problem 2:****Problem 3:**

a)

b)

c)

$$P(\rho_k | x_0) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad (1)$$

$$P(\theta_k | x_0) = \frac{1}{\sigma_b \sqrt{2\pi}} \exp \left\{ -\frac{\left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)\right]^2}{2\sigma_b^2} \right\} \quad (2)$$

Our negative log likelihood function is as follows:

$$\mathcal{L}(x_0; Y_{1:N}) = -\sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)] = J_{NLL} \quad (3)$$

d) In order to minimize (3), we take the gradient with respect to x_0 , or $\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N})$, where

$$\nabla_{x_0} = \begin{bmatrix} \frac{\partial}{\partial \xi_0} \\ \frac{\partial}{\partial \dot{\xi}_0} \\ \frac{\partial}{\partial a_0} \\ \frac{\partial}{\partial \dot{a}_0} \end{bmatrix}$$

$$\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N}) = -\nabla_{x_0} \sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)]$$

It's easy to show that the expression simplifies to the following expression, where the constant terms evaluate to zero.

$$\begin{aligned} &= -\nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu} \right) \right] \\ &\quad - \nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)}{\sigma_b} \right]^2 \right] \end{aligned} \quad (4)$$

Equation (4) consists of two gradient terms to evaluate. Starting with the first term:

$$\nabla_{x_0} \left[\frac{-\nu + 1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right]$$

Since this is the gradient of a scalar term, the output will be a 4-dimensional vector. Each index of the vector is a different partial derivative, and is found to be:

$$\begin{aligned} & \frac{\partial}{\partial \xi_0} \left[\frac{-\nu + 1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\ & \frac{\partial}{\partial \dot{\xi}_0} \left[\frac{-\nu + 1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\ & \frac{\partial}{\partial a_0} \left[\frac{-\nu + 1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\ & \frac{\partial}{\partial \dot{a}_0} \left[\frac{-\nu + 1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \end{aligned}$$

Therefore,

$$\nabla_{x_0} \left[\frac{-\nu + 1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right]$$

$$= \frac{-\nu + 1}{2} \left[\frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right] \quad (5)$$

We will follow a similar process for the second gradient term.

$$\begin{aligned} & \nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)}{\sigma_b} \right]^2 \right] \\ &= \frac{-1}{2\sigma_b^2} \nabla_{x_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ &= \frac{\frac{\partial}{\partial \xi_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ &= \frac{-2 \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right] (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ &= \frac{\frac{\partial}{\partial \xi_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ &= \frac{-2 \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right] (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ &= \frac{\frac{\partial}{\partial a_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ &= \frac{-2 \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)} \end{aligned}$$

$$\begin{aligned}
& \frac{\partial}{\partial \dot{a}_0} \left[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right) \right]^2 \\
&= \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)}
\end{aligned}$$

Therefore,

$$\begin{aligned}
& \nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)}{\sigma_b} \right]^2 \right] \\
&= \frac{-1}{2\sigma_b^2} \left[\begin{aligned} & \frac{-2[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)](a_0 + \dot{a}_0 t_k - 0.5 g t_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ & \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)](a_0 + \dot{a}_0 t_k - 0.5 g t_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ & \frac{-2[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)} \\ & \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)} \end{aligned} \right] \quad (6)
\end{aligned}$$

Finally, we can substitute equations (5) and (6) into equation (4) and we are left with an expression for the gradient vector of $\mathcal{L}_{NLL}$

$$\begin{aligned}
\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N}) = & \sum_{k=1}^N \left[\frac{\nu - 1}{2} \left[\frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2}} \right. \right. \\
& \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2}} \\
& \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(a_0 + a_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2}} \\
& \left. \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(a_0 + a_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + a_0 t - 0.5gt^2)^2}} \right] \\
& + \frac{1}{2\sigma_b^2} \left[\frac{-2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_o + a_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)^2} \right. \\
& \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_o + a_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)^2} \\
& \frac{-2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)]}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)} \\
& \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)]}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)} \left. \right]
\end{aligned} \tag{7}$$

e)

f)